ELCRO FASTENING SYSTEMS

COMPANY PROFILE / ENGLISH

Engineered to connect.

Hook & loop fastening systems
for automotive seat assembly.

HOOK & LOOP / TWIN LOCK

ELCRO CO., LTD.



01 / COMPANY

Fastening expertise, from Ulsan.

Company information and manufacturing location.

COMPANY	DETAIL
Established	February 2017
CEO	Sang Hee Lee
Main products	Hook & loop fasteners
Key clients listed	Hyundai Transys, Hyundai Industrial, Ford
Address	39-1, Hwalcheonsaneop-ro, Duseo-myeon, Ulju-gun, Ulsan
Phone	+82 52 264 9104~5





01 / COMPANY

A record of product development.

Selected milestones from the supplied company history.

FEB 2017	Developed a 1,440 mm-wide mushroom-hook production method.
APR 2017	Supply contract with Beijing DAS, China.
AUG 2019	Supply contract with Ford and Changsung Motors.
AUG 2022	Patent application for the Twin Lock manufacturing method.
JUN 2024	Obtained ISO 9001:2015 quality management certification.
OCT 2024	Artificial-turf application and composite waterproofing development.
JAN 2025	Completed Twin Lock development: mating without distinct male/female parts.
JUL 2025	Twin Lock supply contracts in China and India.

02 / PRODUCTS

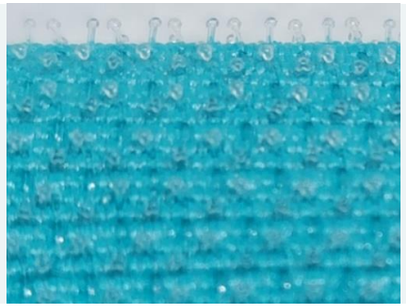
A portfolio of fastening structures.

Textile and plastic hook systems, with textile and automotive loops.



Textile hook

J-shaped hook



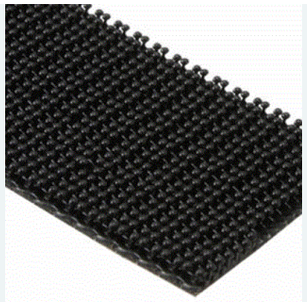
Mushroom hook

Mushroom structure



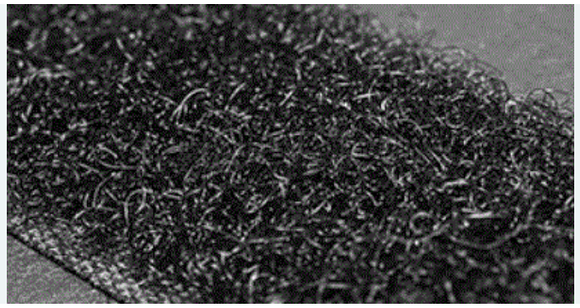
Plastic hook

T-shaped hook



Twin Lock

Self-mating structure



Textile loop

Textile loop surface



Automotive loop

Seat-cover counterpart

03 / APPLICATIONS

Fastening across the vehicle interior.

Seat assembly and interior trim applications shown in the supplied materials.



Seat assembly

Hook & loop fasteners connect the seat pad and seat cover.



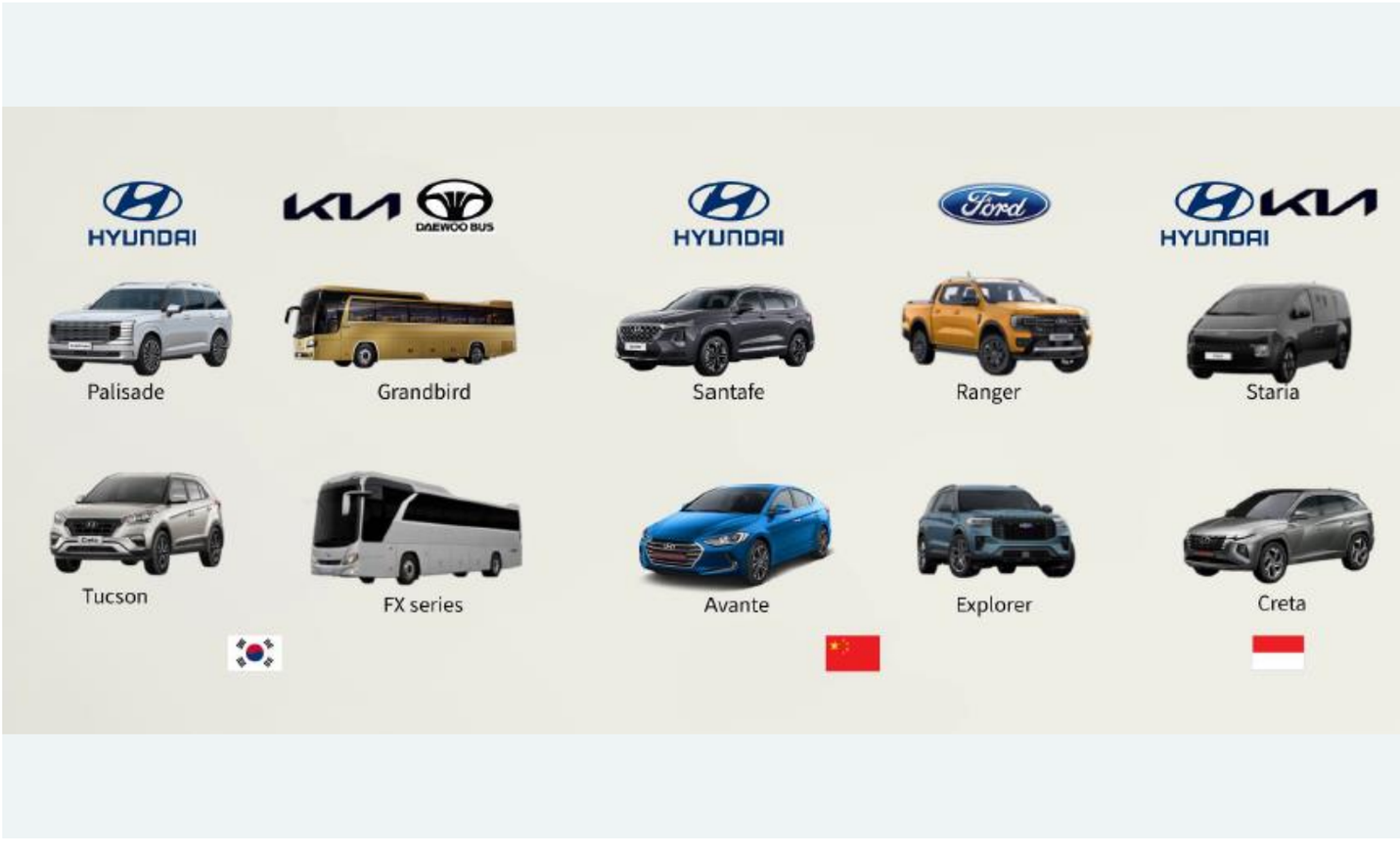
Sunroof & headliner

Twin Lock applications in interior assemblies.

03 / APPLICATIONS

Vehicle applications in the source portfolio.

Illustrative vehicle references; these do not identify the direct purchasing entity.



Referenced models

Hyundai: Palisade, Santa Fe, Staria, Tucson, Avante, Creta

Kia: Grandbird

Ford: Ranger, Explorer

Bus: FX series

Vehicle image and model labels reproduced from the supplied company profile.

04 / HOOK & LOOP

How hook & loop fits into a seat.

Hook in the seat pad. Loop on the seat cover.



HOOK

Fastener integrated with the seat pad.



LOOP

Counterpart positioned on the seat cover.



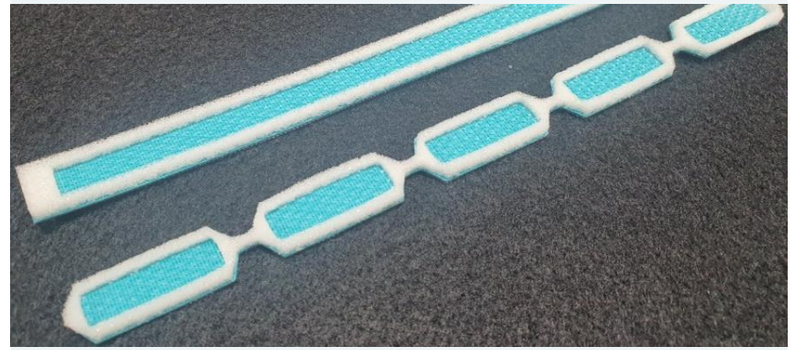
Assembly illustration from the supplied profile.

04 / HOOK & LOOP

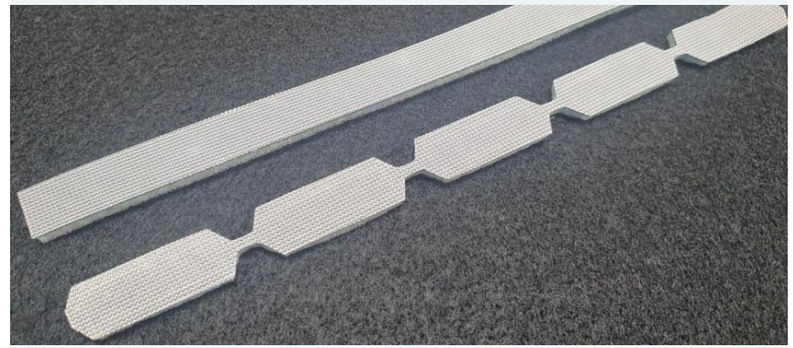
Non-cover hook specifications.

Material stack and nominal dimensions from the supplied specification.

PROPERTY	SPECIFICATION
Composition	Polypropylene mushroom / polyester base / PET film / Fe powder / polyester mesh
Thickness	4.5 mm ± 0.1 mm
Length & width tolerance	± 0.1 mm
Hook & loop separation	15 ± 3 kgf
Hook & seat-pad separation	22 ± 3 kgf
Reference specimen	100 mm (as stated in source)



FRONT



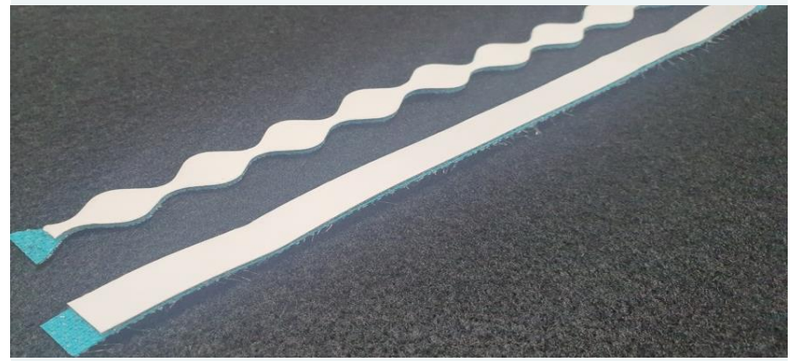
BACK

04 / HOOK & LOOP

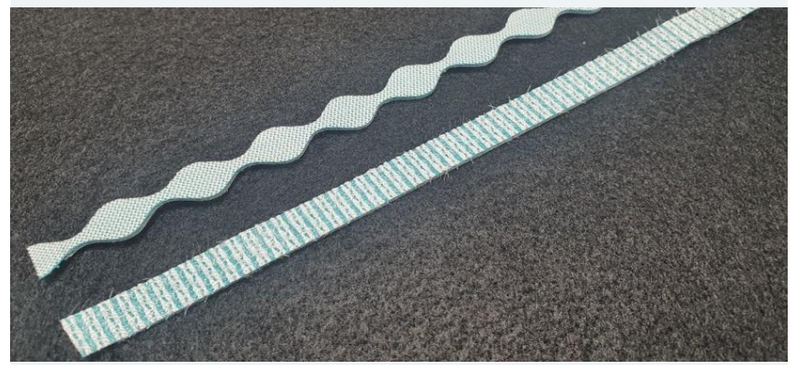
Cover hook specifications.

Covered construction and reported separation-strength requirements.

PROPERTY	SPECIFICATION
Hook composition	Polypropylene mushroom / polyester base / nylon loop
Cover composition	Paper base / Fe powder / nylon (LYLEX)
Thickness	4.5 mm ± 0.1 mm
Length & width tolerance	± 0.1 mm
Separation strength	Hook & loop: 15 ± 3 kgf Hook & seat pad: 22 ± 3 kgf
Reference specimen	100 mm (as stated in source)



FRONT



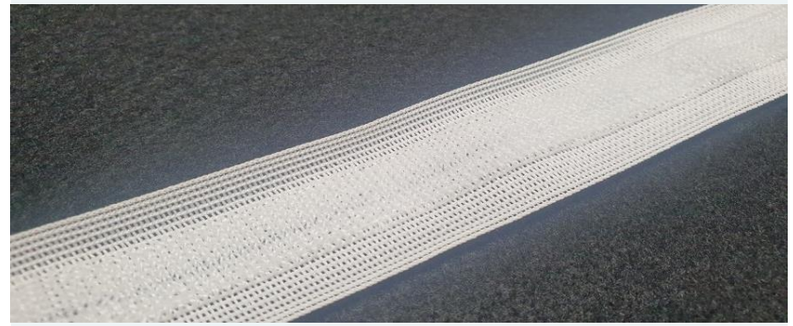
BACK

04 / HOOK & LOOP

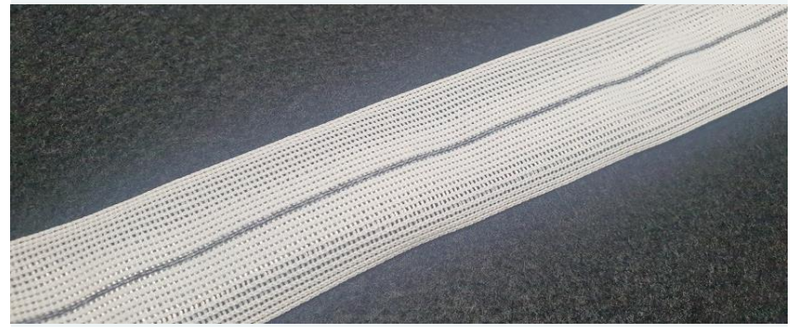
Automotive loop specifications.

Loop material and product dimensions.

PROPERTY	SPECIFICATION
Material	Polyester with phosphorus-based flame-retardant yarn
Thickness	3.2 mm ± 0.1 mm
Available widths listed	40 mm / 38 mm / 36 mm / others
Separation strength	15 ± 2 kgf
Reference specimen	100 mm (as stated in source)



FRONT



BACK

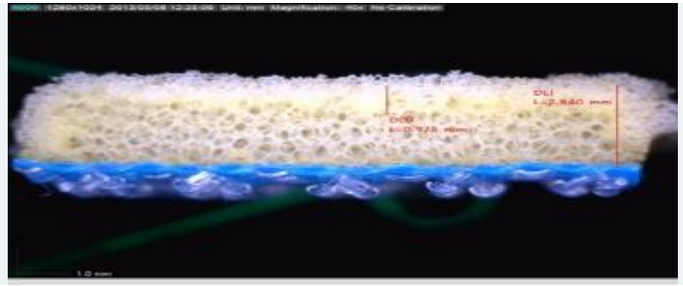
05 / PRODUCT COMPARISON

Integration with the seat pad.

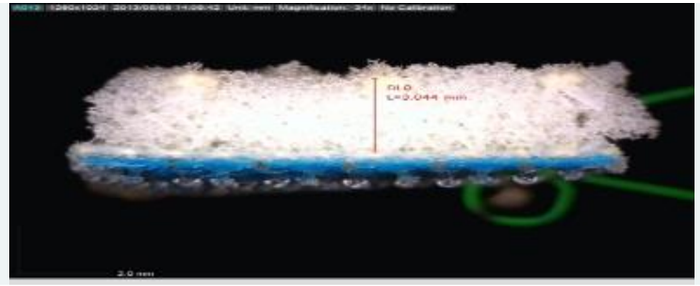
Cross-sections and foaming comparison photographs.

Foam integration

COMPARISON PRODUCT



ELCRO



Source-reported integration: approximately 35 mm for ELCRO versus approximately 1 mm for the comparison product.

Void prevention





Rear nylon-loop structure is described as reducing foaming defects.

Comparisons and performance descriptions are reported in the supplied ELCRO material.

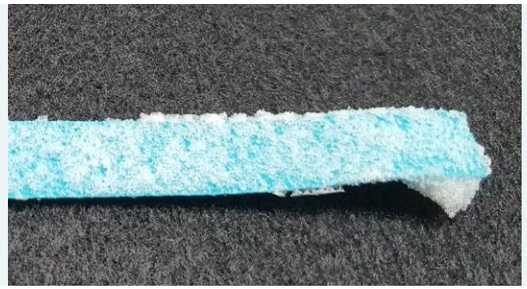
05 / PRODUCT COMPARISON

Bonding and surface consistency.

Rear-layer adhesion and visible surface condition.

Rear-layer bonding

COMPARISON PRODUCT

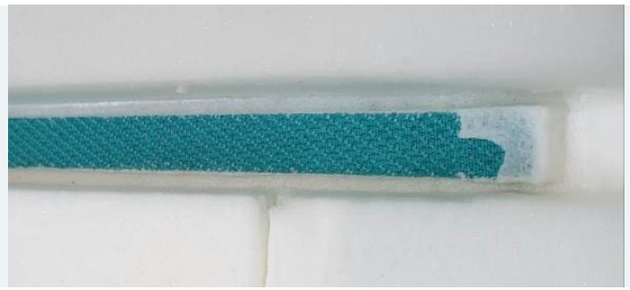


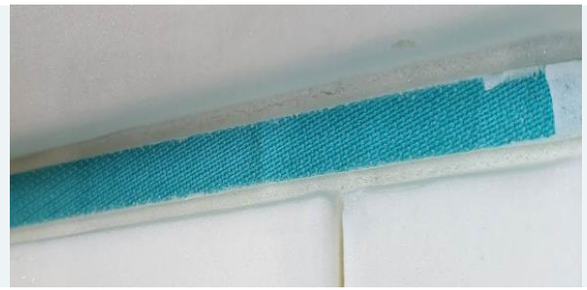
ELCRO



The supplied comparison describes stronger bonding and reduced separation when removing the cover.

Wrinkle comparison





ELCRO is shown without hook wrinkles in the supplied assembly comparison.

Comparisons and performance descriptions are reported in the supplied ELCRO material.

06 / TEST DATA

Measured separation strength.

Source test specimen: 10 mm wide × 100 mm long; vertical separation by P/P gauge.

17 kgf

Hook & loop - reported test result



23 kgf

Hook & seat pad - reported test result



Individual results as provided; specimen geometry and test method are retained.

06 / TEST DATA

Hook reliability tests.

Requirements and outcomes reported in the supplied test summary.

TEST	STANDARD	REQUIREMENT	RESULT
Interior flammability	MS 300-08	Burn rate ≤ 80 mm/min	Approved
Interior odor	MS 300-34	Grade 3 or higher	Approved
Volatile organic compounds	MS 300-55	TVOC < 2 ppm; HCHO < 2 ppm	Approved
Restricted heavy metals	MS 201-02	Pb, Hg, Cd, Cr6+: not detected	Approved







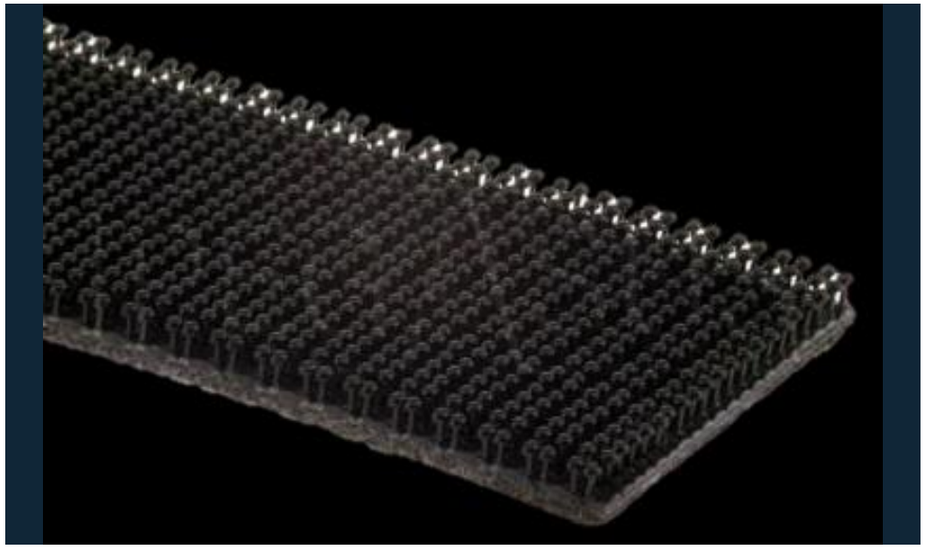


MS refers to Hyundai-Kia engineering standards. Test summary supplied by ELCRO.

07 / TWIN LOCK

Twin Lock. One mating structure.

A self-mating fastener for automotive interior applications.



Self-mating engagement

Matching structures engage without separate male and female fastener parts.



07 / TWIN LOCK

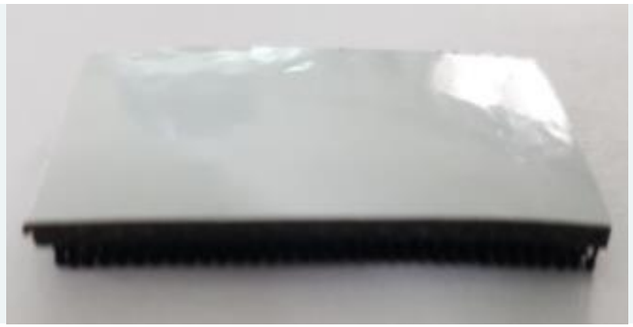
Twin Lock specifications.

Nominal product data and source-reported test values.

PROPERTY	SPECIFICATION
Layer A	Polyolefin
Layer B / C / D	Acrylic foam / acrylic resin adhesive / release paper
Color	Black
Single / combined thickness	3.6 ± 0.2 mm / 5.6 ± 0.3 mm
Protrusions per 25 × 25 mm	400: 20 × 20 stems
Tensile strength	20-28 kgf
Reported reliability	Flame retardancy: 54 mm/min, pass Additional source grade: 1.5, pass Hazardous substances: not detected, pass



FRONT



BACK / ADHESIVE SIDE

08 / TEST DATA

Twin Lock separation across temperatures.

25 × 25 mm specimen. Instron 5582. All values in kgf.

PAIRING	TEMP.	RUN 1	RUN 2	RUN 3	RUN 4
#400 / #400	23°C	25.1	26.8	27.5	25.8
#400 / #400	-30°C	33.2	32.4	35.5	33.7
#400 / #400	120°C	8.3	9.5	8.3	8.7
#400 / #250	23°C	20.8	18.4	19.5	20.2
#400 / #250	-30°C	28.7	25.6	23.4	28.2
#400 / #250	120°C	7.4	7.6	7.1	7.7

Test results supplied by ELCRO. Specimen: 25 × 25 mm. Measurement system: Instron 5582.

08 / TEST DATA

Twin Lock & primer heat resistance.

Source test configuration and reported outcomes.

Test conditions

Chamber: 100°C oven

Loads: 1 kg and 2 kg SPCC steel plates

Substrate: aluminum rail

Primer: 9960 BTX

Test timing: one day after primer application and attachment

Duration: 8 hours

Specimen: 25 × 25 mm

Method: HMC test specification





PAIRING	SOURCE RESULT
#400 / #400	Approved
#400 / #250	Approved

08 / TEST DATA

Twin Lock & primer thermal cycling.

Three cycles; each cycle includes high, low and room-temperature stages.

120°C

8 hours / high temperature

-40°C

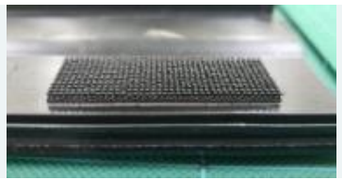
8 hours / low temperature

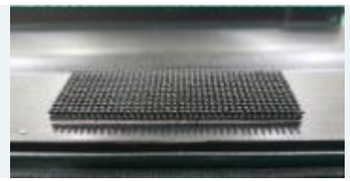
23°C

8 hours / 50% RH

PRIMER	APPLICATION	SOURCE RESULT
520K	Side molding	Pass
4298UV	Side molding	Pass
9960BTX	Rail	Pass







Pairing: #400 / #250. Specimen: 25 × 25 mm. Method: HMC test specification.

08 / TEST DATA

Twin Lock quality test summary.

Automotive interior test standards listed in the supplied profile.

TEST	STANDARD	REQUIREMENT	RESULT
Interior flammability	MS 300-08	Burn rate ≤ 80 mm/min	Pass
Interior odor	MS 300-34	Grade 3 or higher	Pass
Restricted substances	MS 201-02	Prohibition and reporting requirements	Pass





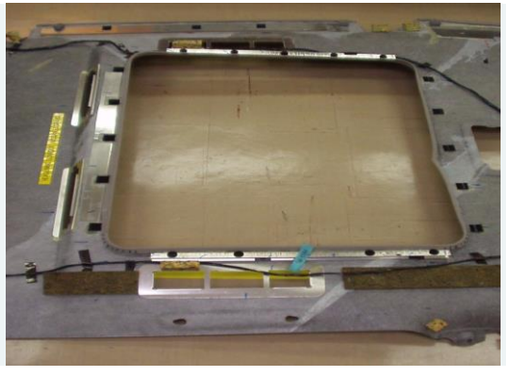


Hyundai-Kia MS standards. Test results and supporting report images supplied by ELCRO.

09 / APPLICATIONS

Twin Lock interior applications.

Application photographs from the supplied ELCRO portfolio.



SUNROOF & HEADLINER



SUNROOF & HEADLINER



QUARTER GLASS

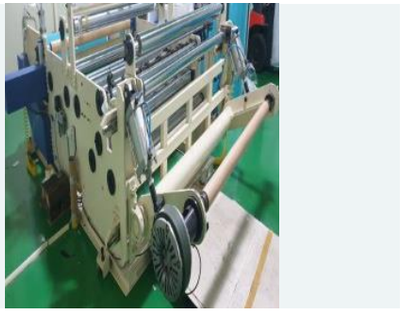


DOOR TRIM

10 / MANUFACTURING

Production equipment.

Equipment categories shown in the supplied manufacturing portfolio.



Thermoforming

Mushroom forming



Slitting

Fabric sizing



Lamination / die cutting

Finished-product sizing



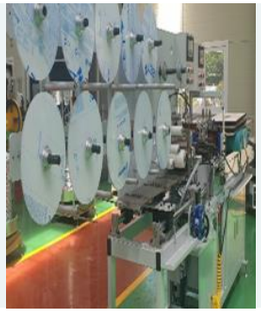
Extrusion coating

Film / fabric coating



Non-cover machine 1

Non-cover production



Non-cover machine 2

Non-cover production



Double forming

Twin Lock molding



Six-fold forming

Twin Lock post-processing

CONTACT

Let’s connect.

Discuss your product requirements and application.

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getelcro.com